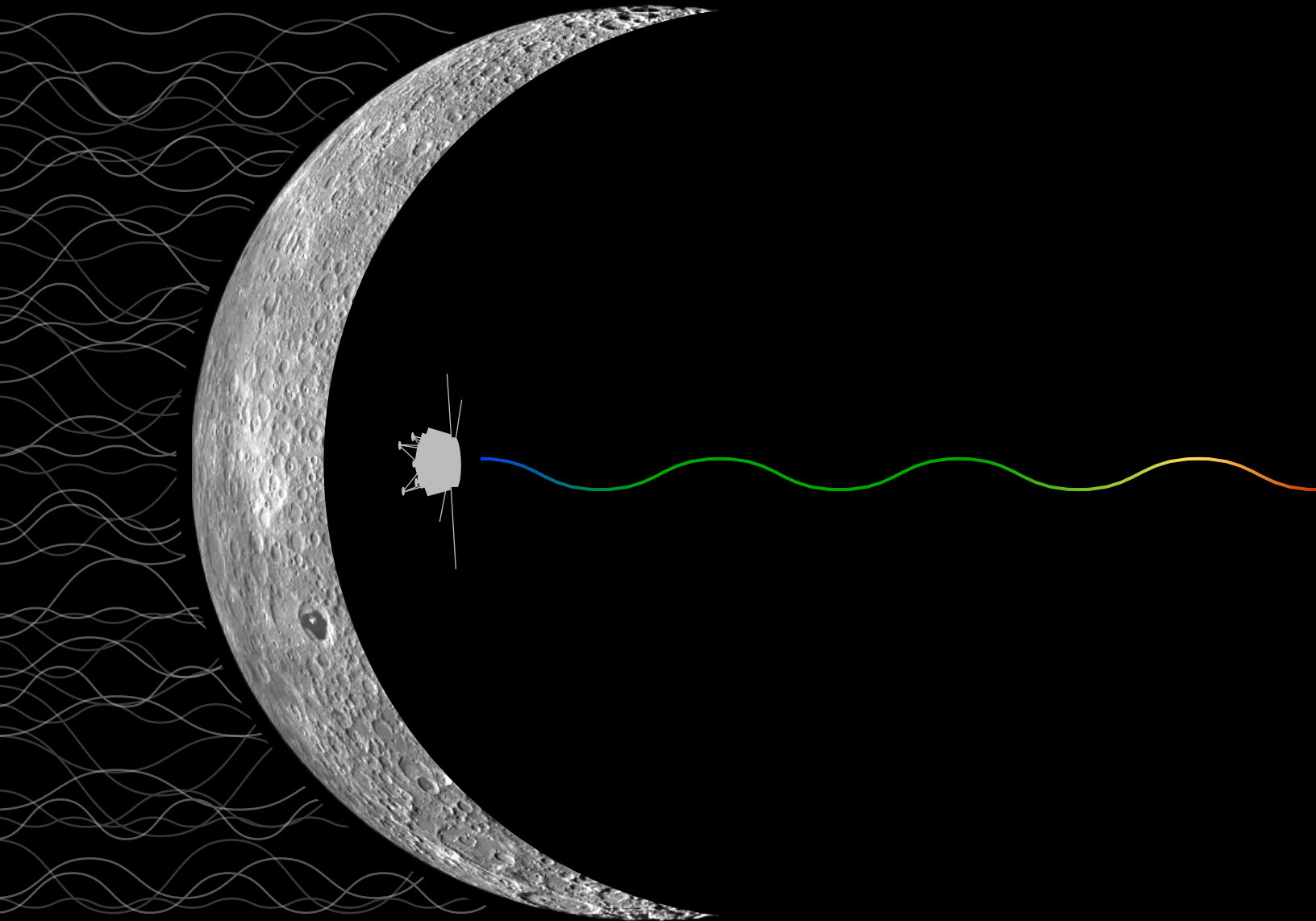


LFT3

Lunar Farside Transients and Technology Telescope

A Legacy Survey Mission



BREAKTHROUGH
LISTEN

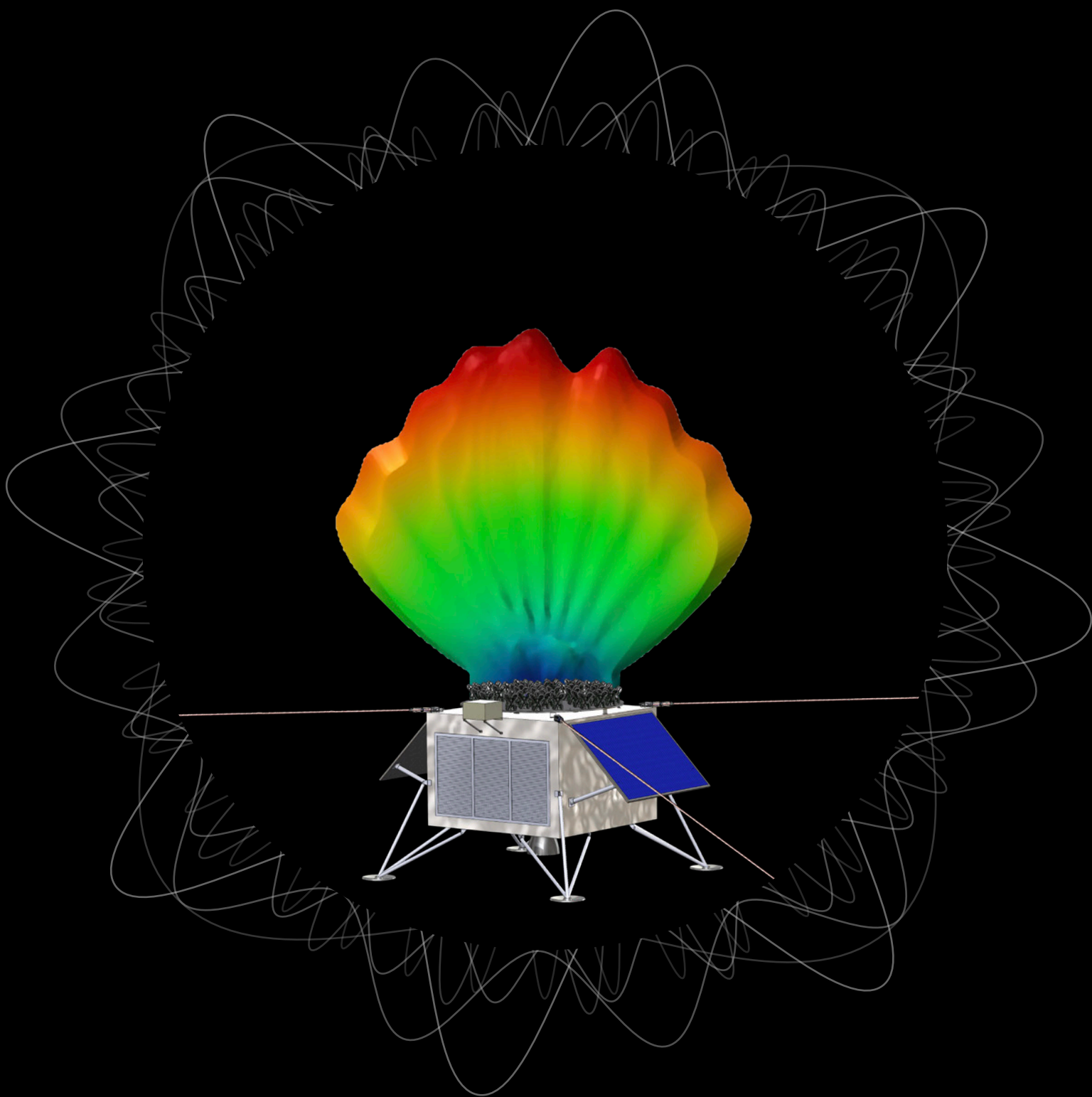
BREAKTHROUGH
INITIATIVES

UNIVERSITY OF
OXFORD

RADIO ASTRONOMY
SYSTEMS | BYU

SSL

HELIOSPACE





Radio Science Impact

Radio wavelengths provide the ability to observe and understand the roughly 90% of ordinary matter that comprises our Universe and is inaccessible to other wavelength bands. Radio wavelengths also uniquely allow us to observe and survey the entirety of the very early times of the Universe when the first structures began to coalesce.

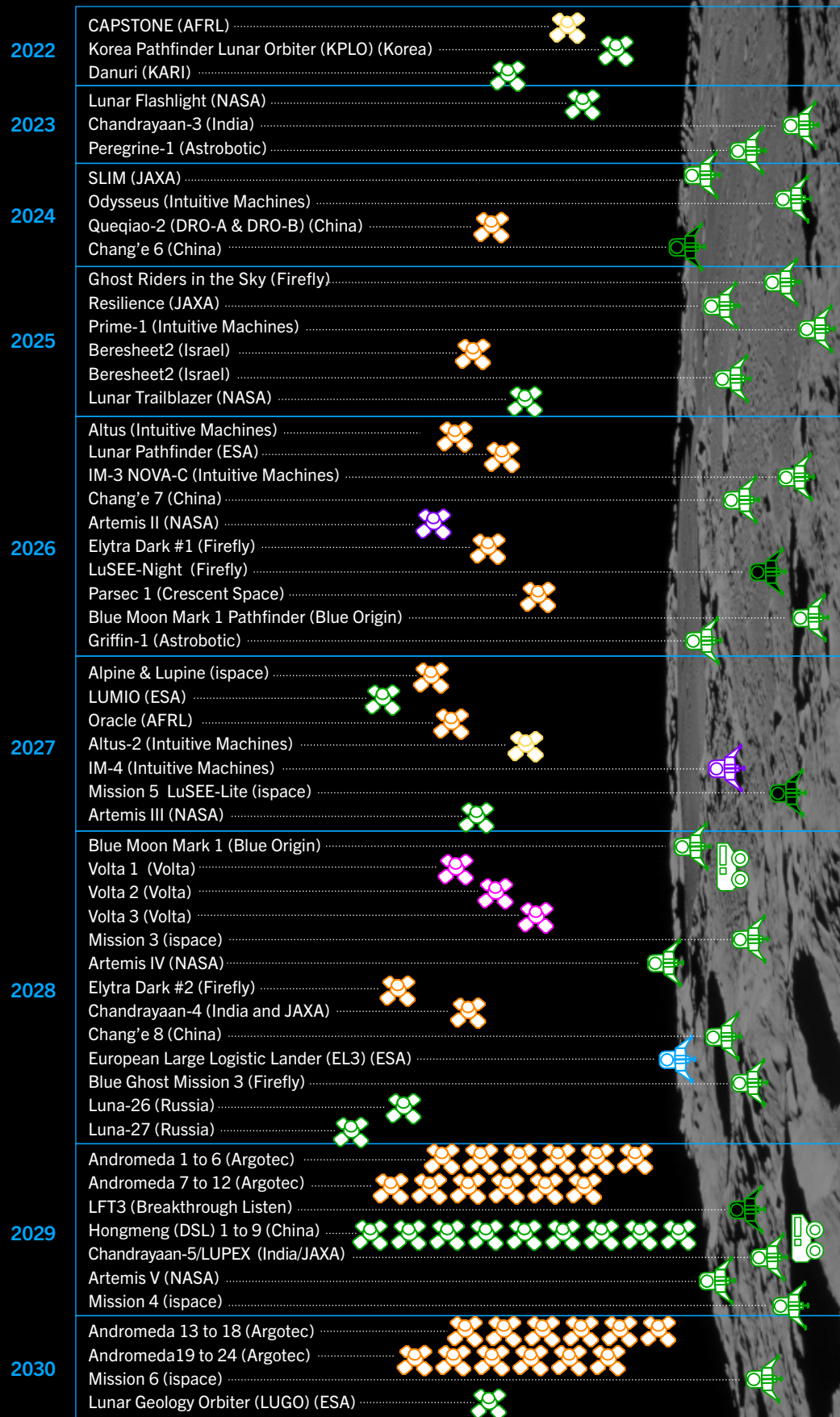
Radio wavelengths potentially provide our best chance to glimpse technological signals from other civilizations in our Universe.

The lunar farside is a uniquely radio-quiet location in the inner solar system since it is always blocked from Earth's line-of-sight.

Humanity is poised to explore the cislunar environment, but has not yet arrived in numbers.

Now is the time to conduct this once-in-human-history survey of the quiet cosmic radio frequency environment.

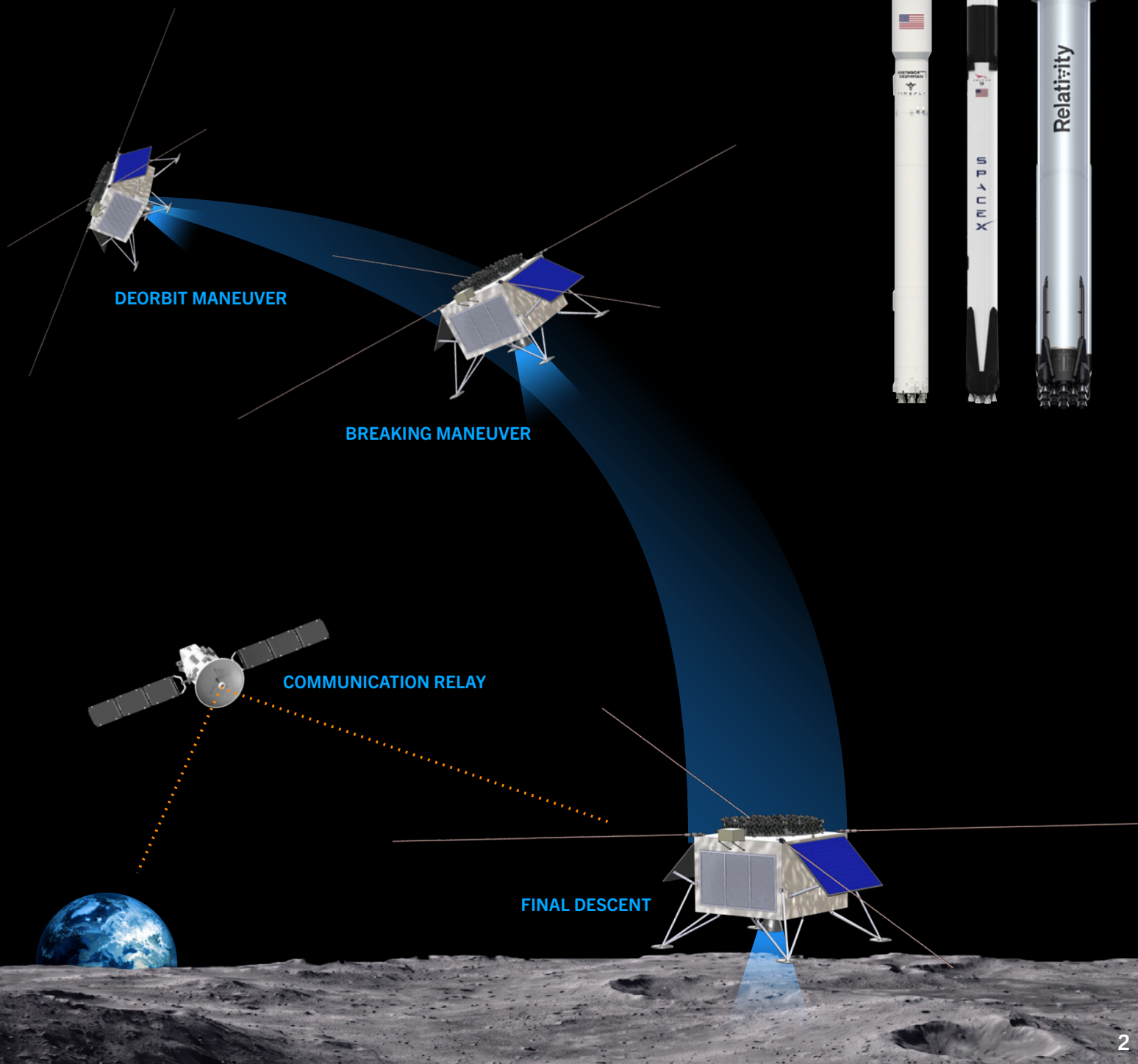
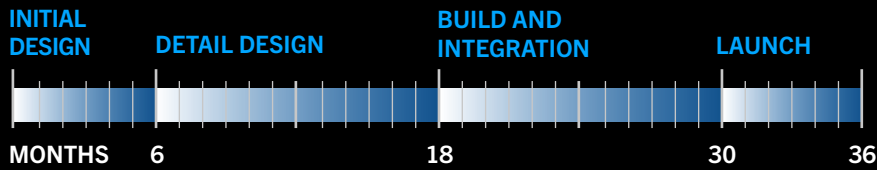
By the end of this decade, many orbiters and landers will be present, impacting the pristine environment and humanity will lose this unique opportunity.





Getting to the Moon

Commercial Launch Vehicle, Lunar Lander, and Communication Relay vendors exist that have the capability to execute the LFT3 mission.





The 1973 measurements taken by NASA's RAE-2 mission highlighting the extreme radio-quiet lunar farside environment remain the exemplar, however recent missions (like China's 2018 Longjiang orbiters) are beginning to show this same promise.

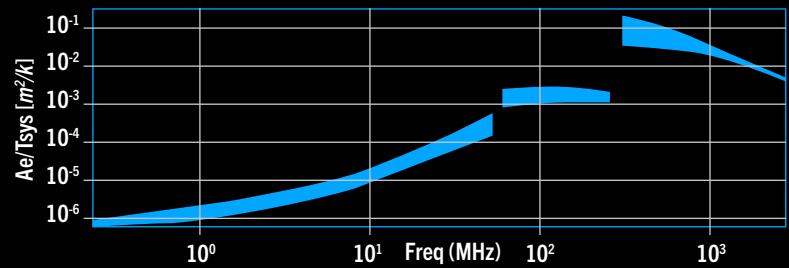
LFT3 builds on the NASA/DOE LuSEE-Night stand alone pathfinder in 2026 to land on the farside and use lander resources to provide continuous far side observations for 6 Earth years.

China is the only country to land and operate from the lunar farside with their Chang'E 4 mission, which was corrupted by self-interference.

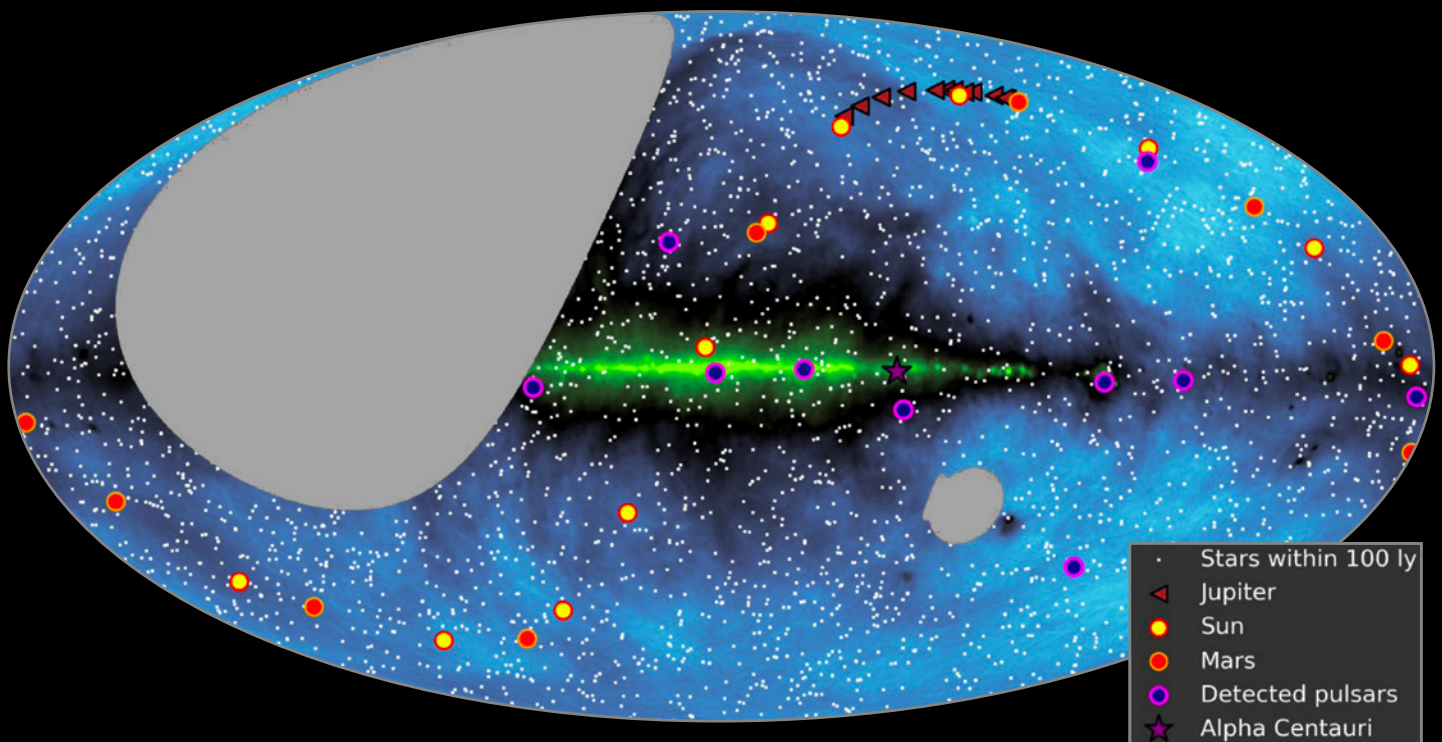
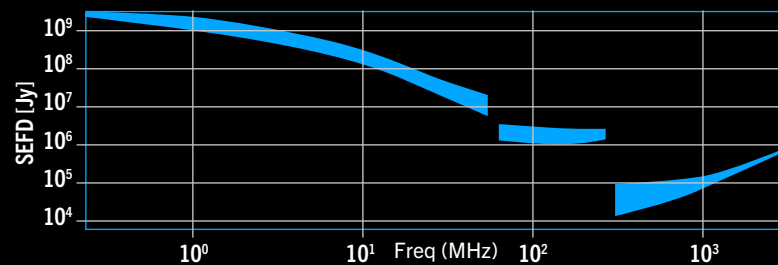
The full system provides sensitive frequency coverage from 1-2700 MHz.

The LFT3 midband compact dual polarization multi-beam phased array provides wide sky coverage with full aperture sensitivity.

EFFECTIVE AREA / SYSTEM TEMPERATURE



SOURCE EQUIVALENT FLUX DENSITY





Study unique periodic or transient events across a wide range of wavelengths and timescales.

Search for extraterrestrial life beyond the sphere of human influence and across the radio band.

Observe stellar dynamic nature at high spectral resolution.

Explore new physics at frequencies below the ionospheric barrier and across cosmic time.

Establish incumbent cislunar scientific use of frequencies across the microwave spectrum.

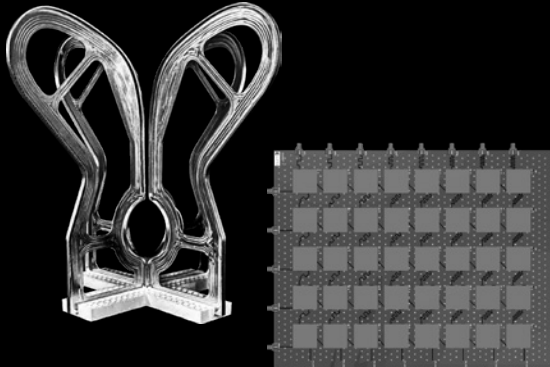
Mission Objective	Receiver Band	Freq. Res.
Pulsar	HF/VHF/UHF	0.5MHz
Fast Radio Bursts Analogs	HF/VHF/UHF	0.5MHz
Technosignatures	HF/VHF/UHF	10Hz
Flare Stars	HF/VHF/UHF	10 MHz
Exoplanets	HF	10MHz
Solar System	HF	100 kHz
	HF	200 kHz
	HF	10 MHz
Studies of the Sun	HF/VHF	0.5MHz
	VHF	0.5MHz
Hydrogen Line Studies	HF	100–500Hz
	VHF/UHF	1kHz
Radio Recombination Lines	HF	100–300Hz
	VHF/UHF	100–300Hz
Cosmology	HF/VHF	1 MHz
Very Long Baseline Interferometry	VHF/UHF	
Lunar studies	HF/VHF/UHF	1 kHz

LFT Science Goals

LUNAR-EXCLUSIVE SCIENCE	LUNAR-AUGMENTED SCIENCE	PROOF-OF-CONCEPT SCIENCE
Novel studies uniquely enabled by the lunar farside	Enhance and extend Earth-based observations	Demonstrate capability through proven science
TRANSIENTS Discover local low-DM transients in the FM band and below the ionospheric cutoff	FRBs Perform wideband observations of repeating FRBs at low frequencies	SOLAR PHYSICS Perform high-cadence, wideband observations of solar flares
COSMOLOGY Capture sky-complete spectra in HF and VHF bands	PULSARS Perform wideband observations of pulsars	FRBs Continue FRB discovery through widefield UHF monitoring
TECHNOSIGNATURES Perform RFI-free technosignature surveys in frequency bands (HF) inaccessible from Earth	TECHNOSIGNATURES Observe known anthropogenic signals from Mars across all LFT3 frequencies	RRL Perform spectral measurements of galactic and extragalactic radio recombination lines
LUNAR RF-MONITORING Perform legacy survey of the lunar farside across all LFT3 frequencies (1 MHz – 2.7 GHz)	HYDROGEN LINES (HI) Complement ground surveys of extra-galactic HI with RFI-free, stable spectral observations	COSMOLOGY Capture sky-complete spectra in VHF and higher frequencies
SOLAR SYSTEM EMISSION Observe Jupiter HOM/DAM, Saturn SKR, Uranus UKR emissions and planetary lightning	SOLAR PHYSICS Observe solar activities at frequencies (HF) that have not been explored since Voyager	
EXOPLANETS Search for exoplanet magnetospheric emission in the space-only 0.1 – 10 MHz band	EXOPLANETS Extend ground-based surveys to lower-frequency emission from different planet populations	



Primary Instrumentation



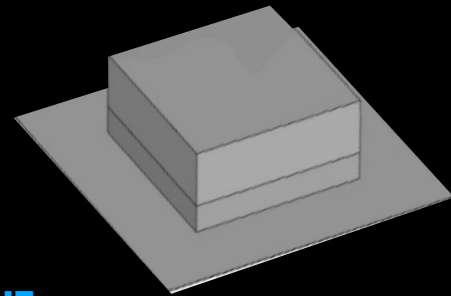
UHF

The LFT UHF compact dual polarization multi beam Vivaldi phased array provides wide sky coverage with full aperture sensitivity. Operating bandwidth is 300 - 2700 MHz. The system incorporates a Blass matrix analog beamformer to produce 10 dual-polarization beams.



Spectrometer

The flexible spectrometer allows for on-board dynamic spectra to be computed and stored along with triggered baseband data. It interfaces with the rest of the system to provide data to be transmitted back to Earth.



VHF

The VHF antenna, operating in the 60 - 260 MHz range, will incorporate meta materials to increase bandwidth and gain.



HF Antennas

The LFT3 HF Antennas operating in the 0.1 to 50 MHz range and are the same as those flown on LuSEE-Night and have heritage to the antennas for ROLSES.



Agentic On-Board

On-board edge intelligence allows for local programmatic decisions to optimize the science return of the mission. LFT3 will be the first mission to deploy AI on the moon.



Overall program to deploy autonomous telescopes across the globe for continual searching, including time-of-arrival discrimination.

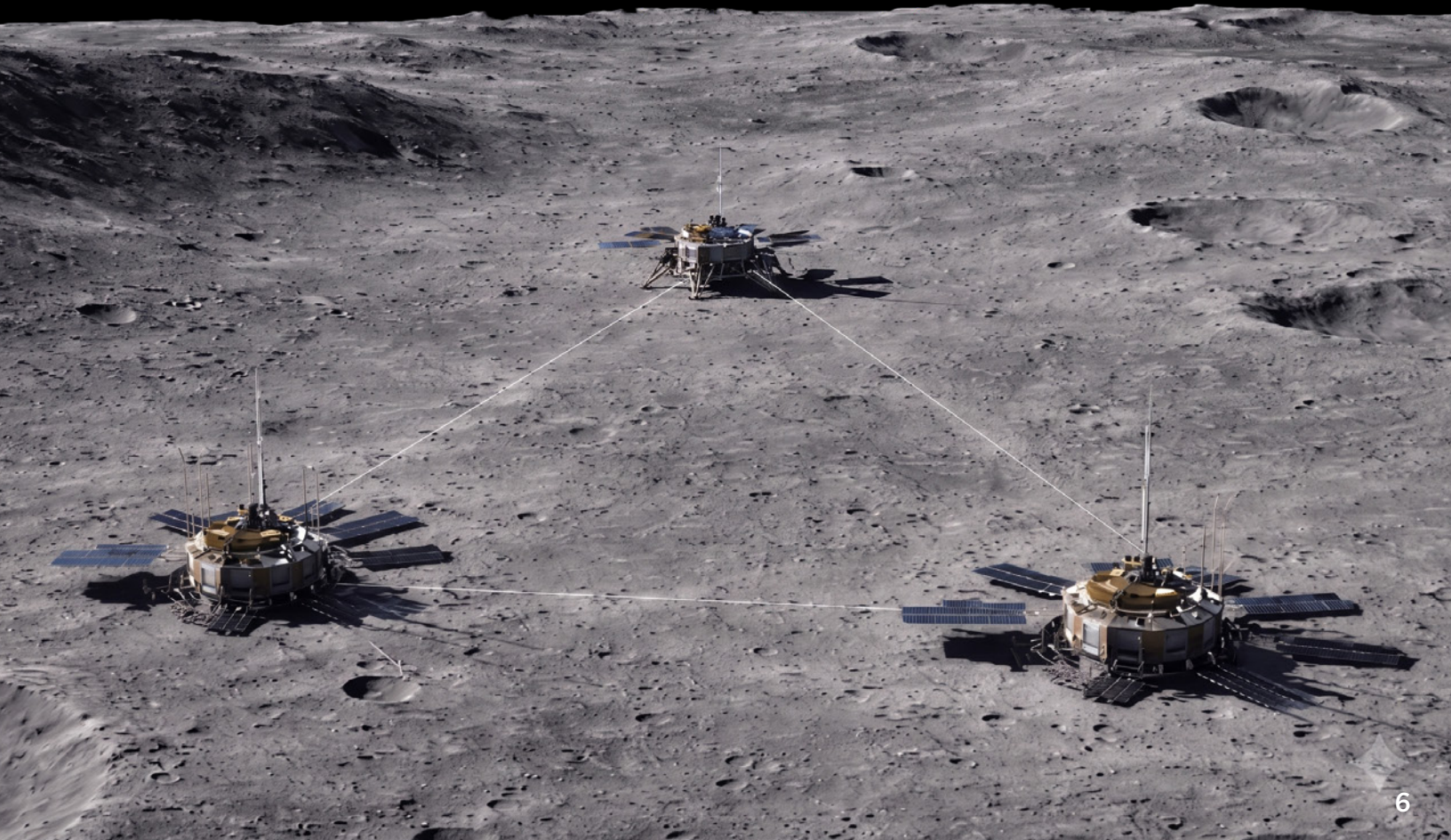
Agentic AI will be used to optimize the science return. Earth-based systems can trial and extend space-based techniques.

Extend the network to the moon with multiple installations for sensitivity and time-of-arrival localization.

Long-standing collaboration, including NVIDIA, to develop agentic observatory concepts for earth and space applications.



LFT3 system in Bhutan (rendering)





Science Traceability Matrix

GOALS AND OBJECTIVES	MEASUREMENTS REQUIRED	REQUIRED CAPABILITIES
TRANSIENTS Study the low-frequency emission from transients, pulsars, FRBs, and technosignatures in an RFI free environment and at frequencies not accessible from Earth	Discovery-driven surveys for low-frequency radio transients	H F
	Search for low-DM transients	U M V
	Make low-freq observations of known pulsars to measure their low frequency spectrum	H F
	Make wideband observations of known FRB repeaters	H U F V
	Unambiguous blind search for technosignatures in all LFT3 bands	N H U M V
COSMOLOGY AND SPECTRAL LINES Explore the early and local Universe through spectral lines of interest	Make spectral line observations of galactic and extragalactic sources at frequencies of interest (H1, RRL, masers, etc)	N U M V
	Measure redshifted 21 cm absorption spectra from the Early Universe	H V
SOLAR SYSTEM Investigate sources of radio emission from within the solar system	Perform wideband high cadence observations of solar transient events (e.g. flares, CMBs)	H U F V
	Study and validate the low-frequency auroral emissions from the outer planets and lightning emissions from all the planets	H F
EXOPLANETS Investigate radio emission from nearby exoplanets	Search for radio emissions from exoplanets in the unexplored low-frequency spectrum (<10 MHz)	H
RF ENVIRONMENTAL STUDIES OF MOON Perform legacy surveys of the Moon's RF environment	Measure the low frequency (<300 MHz) spectra of radio-bright stars and ultra cool dwarfs, in the unexplored low-frequency spectrum (<10 MHz)	H V
RADIO-BRIGHT STARS Characterise the low-frequency spectra of radio-bright stars and ultra-cool dwarfs	Perform baseline RF environmental surveys of the far side of the Moon	N H U V
LOW-FREQUENCY VLBI Establish a new observational framework for VLBI	Demonstration of new high-resolution low-frequency VLBI observations alongside Earth-based observatories	H U V

N Narrow mode (<1kHz)
 U UHF Band 300 - 2700 MHz
 M Multibeam (~10)
H HF Band 0.1 - 50 MHz
 F Fast mode (<1s)
 V VHF Band 60 - 260 MHz



For more information contact David DeBoer
dave.deboer@physics.ox.ac.uk

